BIOHYDRA-ODISHA

Waste-to-Hydrogen Circular Bioenergy System

Turning Agricultural Waste into the Energy of Tomorrow

An innovation concept for converting suitable sugarcane bagasse into biological hydrogen through two-stage lactate-driven dark fermentation.

EXECUTIVE SUMMARY

BioHydra-Odisha proposes a modular biological process to convert sugarcane bagasse—the fibrous residue left after sugarcane juice extraction—into hydrogen-rich gas. The concept combines biomass pretreatment, two-stage lactate-driven dark fermentation, sensor-based process monitoring, gas purification and responsible recovery of residual materials. The project would be developed through laboratory validation, a small pilot plant and, if successful, an industrial demonstration.

Core value: Integrate waste valorization, biotechnology and smart process control in a locally adaptable system, while measuring real hydrogen yield, purity, cost and environmental performance.

CONTEST PROPOSAL STRUCTURE

- **1. Title:** BioHydra-Odisha — Waste-to-Hydrogen Circular Bioenergy System
- **2. Brief idea:** Convert suitable bagasse into biological hydrogen using a two-stage fermentation process.
- **3. Core technology:** Pretreatment, microbial fermentation, smart monitoring, gas purification and resource recovery.
- **4. Uniqueness:** A locally adapted, modular system with measurable performance and lifecycle evaluation.

1. Problem Statement

Sugarcane processing generates bagasse, a lignocellulosic residue containing cellulose, hemicellulose and lignin. Bagasse already has established uses, including as boiler fuel, so the proposal does not assume that all bagasse is waste. Instead, it targets suitable surplus or underutilized material and investigates whether biological hydrogen production can provide additional value.

Proposed solution

Develop a modular, two-stage lactate-driven dark-fermentation system. In the first reactor, selected microorganisms convert accessible sugars into lactate-rich intermediates. In the second reactor, a compatible hydrogen-producing microbial culture converts suitable intermediates into hydrogen. The resulting gas is measured, cleaned and tested before any use.

2. How the Process Works

Stage	Process	Purpose
1. Collection	Collect and inspect bagasse; remove soil and contaminants.	Secure a consistent feedstock.
2. Pretreatment	Shred biomass and test suitable physical, chemical or biological pretreatment.	Make carbohydrates more accessible.
3. Sugar release	Use suitable enzymes or hydrolysis methods to release fermentable sugars.	Prepare substrate for fermentation.
4. First reactor	Operate an oxygen-free lactate-producing fermentation stage.	Create suitable intermediates.
5. Second reactor	Use selected hydrogen-producing microorganisms under optimized conditions.	Generate hydrogen-rich gas.
6. Purification	Remove moisture and other impurities using an appropriate separation train.	Meet verified gas-quality requirements.
7. Recovery	Assess residual liquid and solids for further energy/material recovery and safe treatment.	Improve overall resource efficiency.

Important: This is a proposed process flow, not a claim that the full system has already been built or validated.

3. Core Technology and Methodology

A. Lignocellulosic pretreatment

Bagasse's fibrous structure limits direct microbial access to its carbohydrates. Compare practical pretreatment options for sugar recovery, chemical use, water consumption, cost and wastewater generation.

B. Two-stage lactate-driven dark fermentation

Separate intermediate production from hydrogen generation so each stage can be optimized independently. Lactate accumulation does not by itself guarantee hydrogen production; compatible cultures and operating conditions must be verified experimentally.

C. Smart bioreactor monitoring

Use sensors for pH, temperature, pressure and gas flow, connected to a microcontroller or PLC for data logging, alerts and process control. Monitoring supports consistency but does not replace laboratory gas analysis.

D. Hydrogen purification

Choose purification equipment based on measured gas composition and flow. Pressure swing adsorption (PSA) may be evaluated, but suitability and achievable purity must be demonstrated. A purity such as 99.999% is a target to verify, not a guaranteed outcome.

E. Circular resource recovery

Characterize residual solids and liquids, evaluate potential recovery of energy or useful materials, reuse water where safe, and treat wastewater to meet applicable requirements.

Development methodology

- Characterize bagasse moisture, total solids, volatile solids, carbohydrate and lignin content.
- Run controlled pretreatment trials and measure fermentable-sugar recovery.
- Select compatible microbial cultures with a qualified biotechnology laboratory.
- Optimize pH, temperature, substrate loading, retention time and microbial balance.
- Measure gas volume and composition with calibrated instruments; quantify hydrogen separately from carbon dioxide, methane and other gases.
- Test purification performance, leakage controls and gas quality.
- Compare results with an appropriate baseline using the same feedstock basis.
- Complete techno-economic and lifecycle assessments, including electricity, water, chemicals, transport and waste treatment.

4. Uniqueness and Advantages

Challenge in existing approaches	BioHydra-Odisha improvement to test
Bagasse may be underutilized, but it also has existing uses.	Target suitable surplus material rather than diverting all bagasse from current uses.
Single-stage fermentation may be affected by competing pathways.	Separate intermediate formation and hydrogen production for independent optimization.
Pretreatment can add cost and environmental burden.	Compare options using sugar recovery, cost, water, energy and chemical use.
Manual operation can make performance inconsistent.	Use sensors, data logging and alerts to detect operating deviations.
Residual streams may create additional waste-treatment needs.	Assess safe by-product recovery and integrate wastewater management.
Large facilities may not suit every feedstock location.	Develop a modular pilot design adaptable to suitable processing sites.
Clean-energy claims can overlook upstream impacts.	Measure lifecycle emissions, energy use and cost per kilogram of hydrogen.

EVIDENCE FOR ODISHA'S LOCAL POTENTIAL

Published findings support feasibility and justify local testing — they do not yet prove that this exact proposed system performs better in Odisha.

What the evidence shows

Evidence	Key finding	Relevance to BioHydra-Odisha
Odisha regional context — APICOL, "Bagasse based Unit"	An Odisha government project document identifies sugarcane cultivation in several historical undivided districts and notes existing bagasse uses such as fuel, compost and paper.	Supports regional relevance, while showing the project should target suitable surplus bagasse rather than assume all bagasse is waste.
Kumari & Das, *Bioresource Technology* (2015)	In a specific lab study, alkali pretreatment improved enzymatic hydrolysis about 2.6-fold; a two-stage hydrogen-plus-methane process reported 44.8% energy-conversion efficiency versus 5.4% for single-stage H■-only processing.	Supports testing pretreatment and staged conversion. These figures are study-specific and are not predicted results for this design.
Jain & Panwar, *Journal of Agricultural Engineering* (2026; OUAT affiliation)	A two-stage bagasse fermentation study reported maximum yields of 103.9 mL H■/g total solids and 314.4 mL biomethane/g total solids under its optimized, pretreated conditions.	Provides locally connected research evidence that pretreated bagasse can be investigated through two-stage fermentation.
Deka et al., *ChemistryOpen* (2025)	A modelling study estimated higher Indian agricultural-residue biohydrogen potential with pretreatment than without it (about 300 PJ vs 103 PJ across seven residues).	Supports the broader rationale for pretreatment, but this is a national model, not measured Odisha output.

How to demonstrate performance under Odisha conditions

Run a controlled comparison using the same locally sourced bagasse: (A) untreated feedstock, (B) pretreated feedstock, and (C) pretreated feedstock with the proposed two-stage lactate-driven process. Use repeated trials and compare hydrogen yield per unit of dry feedstock, gas composition/purity, processing time, water, electricity and chemical use, process stability, and estimated cost. Work with a qualified laboratory and appropriate hydrogen-safety controls.

Evidence-based conclusion: Existing research supports the feasibility of pretreating bagasse and investigating staged fermentation. Odisha has a relevant agricultural context and local research links, but superiority of the specific BioHydra-Odisha configuration must be established through experiments with local feedstock.

Important qualification: Published values come from different experimental systems and measurement bases, so they should not be compared as if they were results from one common test. No claim of guaranteed yield, purity, cost advantage or Odisha-specific superiority is made.

5. Implementation Roadmap

Phase 1 — Laboratory validation

Test feedstock pretreatment and fermentation in controlled laboratory reactors. Establish repeatable yield, gas composition and operating conditions.

Phase 2 — Small pilot plant

Integrate the two reactors, sensors, gas measurement and an appropriate purification unit. Test real bagasse and evaluate process stability, safety and operating costs.

Phase 3 — Industrial demonstration

If the pilot is successful, partner with sugar-processing facilities, research institutions and clean-energy organizations to assess scale-up and commercial feasibility.

6. Success Indicators

Indicator	Measurement
Hydrogen yield	Volume or mass of hydrogen per kilogram of dry bagasse or volatile solids; state the basis clearly.
Hydrogen purity	Hydrogen concentration and verified impurity levels.
Process stability	Variation in yield and operating conditions across repeated runs.
Resource efficiency	Water, electricity and chemical use per kilogram of hydrogen.
Economic feasibility	Estimated total production cost per kilogram of hydrogen.
Environmental impact	Lifecycle greenhouse-gas emissions compared with a defined baseline.
Circularity	Amount of residual material safely recovered or reused.

7. Expected Benefits for Odisha

- **Waste valorization:** Create an additional use for suitable surplus bagasse.
- **Clean-energy research:** Explore a biological pathway that complements other hydrogen production methods.
- **Local opportunities:** Potential work in feedstock collection, plant operation, maintenance and testing if the process becomes viable.
- **Circular bioeconomy:** Investigate recovery of useful materials and responsible management of residual streams.

8. Final Pitch

BioHydra-Odisha is a proposed waste-to-energy innovation that aims to convert suitable sugarcane bagasse into hydrogen through two-stage lactate-driven dark fermentation. By combining optimized biomass pretreatment, selected microbial cultures, sensor-based reactor monitoring and suitable gas purification, the project seeks to build a modular biological hydrogen system adapted to Odisha's agricultural and industrial ecosystem. The project will progress from laboratory validation to pilot testing, measuring hydrogen yield, purity, process stability, production cost and lifecycle emissions. Its long-term vision is to create additional value from agricultural residues while supporting rural opportunities, circular resource use and India's transition towards cleaner energy.

Vision: To transform suitable agricultural residues into a locally valuable energy resource through scientifically validated, scalable and environmentally responsible biotechnology.